

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fourth Semester B.Tech Degree Examination July 2021 (2019 Scheme)

ANSWER KEY**Course Code: EST200****Course Name: DESIGN AND ENGINEERING**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions. Each question carries 3 marks*

Marks

1	Outline the significance of understanding customer requirements in design process	(3)
Ans	Design requirements state the important characteristics that your solution must meet to succeed. One of the best ways to identify the design requirements for your solution is to analyze the concrete example of a similar, existing product, noting each of its key features. • Design requirements state the important characteristics that your design must meet in order to be successful. • One of the best ways to identify the design requirements for your solution is to use the concrete example of a similar, existing product, noting each of its key features. Here is how to analyze: ○ A physical product ○ A software product or website ○ An environment ○ An experience • To complete the requirements step of the design process, you should write a design brief; a document that holds all of the key information for solving your problem in one place	
2	Describe any three constraints that can occur in design process of a lunch box	(3)
Ans	Design constraints . 1. Easy to handle and portable . 2. Hybrid slot . 3. Light weight and easily carried . 4. Ecofriendly material The lunch box has most often been used by schoolchildren to take packed lunches, or a snack, from home to school. The most common modern form is a small case with a clasp and handle, often printed with a colorful image that can either be generic or based on children's television shows or films	
3	Explain convergent questioning in design thinking	(3)
Ans	Convergent and divergent thinking Divergent thinking • It is a thought process or method used to generate creative ideas by exploring many possible solutions. • It typically occurs in a spontaneous, free-flowing, "non-linear" manner, such	

	that many ideas are generated in an emergent cognitive fashion. • Divergent thinking uses the imagination to open the mind to new possibilities and solutions, and ultimately become more innovative. Convergent thinking • It is the opposite of divergent thinking. • It generally means the ability to give the "correct" answer to standard questions that do not require significant creativity, for instance in most tasks in school and on standardized multiple-choice tests for intelligence. • Convergent thinking moves from broad thoughts to concrete understanding, where the thoughts from divergent thinking can be narrowed down to the most promising ideas and solutions.	
4	Explain how conflict in team environment helps in better design of products	(3)
Ans	• Members of a Design Thinking team need to be open minded, curious, collaborative and allow their assumptions to be challenged, ready for change, and be adaptable. • Cross-disciplinary teams will provide you with the best results. • Teams may consist of people unfamiliar with each other, with external members brought on board either as specialists or facilitators depending on the availability of skills. • To make a Design Thinking project successful, we need T-shaped people. • T-shaped people have a depth of knowledge and experience in their own fields but they can also reach out and connect with others horizontally and create meaningful collaborations. • All team members should be encouraged to respect each other's inputs, in order to discover deeper and to build upon each other's findings.	
5	Describe how prototyping helps in design process	(3)
Ans	Prototyping is an experimental phase. • The aim is to identify the best possible solution for each problem found. • Prototypes may be shared and tested within the team itself, in other departments, or on a small group of people outside the design team. • Team should produce some inexpensive, scaled-down versions of the product (or specific features found within the product) to investigate the ideas you've generated. • The solutions are implemented within the prototypes, and, one by one, they are investigated and either accepted, improved and re-examined, or rejected on the basis of the users' experiences. • By the end of this stage, the design team will have a better idea of the constraints inherent to the product and the problems that are present, and have a clearer view of how real users would behave, think, and feel when interacting with the end product.	
6	Explain any three advantages of communicating designs in writing.	(3)
Ans	• REPORTING is an essential part of a design project • We communicate final design results in several ways, including oral	

	presentations, final reports (that may include design drawings and/or fabrication specifications), and prototypes and models. • The primary purpose of such communication is to inform our client about the design, including explanations of how and why this design was chosen over competing design alternatives. • It is most important that we convey the results of the design process. • We should ensure that final reports and presentations are not narratives or chronologies of our work. Rather, the presentations and reports should be lucid descriptions of design outcomes, as well as the processes with which those outcomes were achieved.	
7	Illustrate advantages of reverse engineering in design	(3)
Ans	1. Exploring existing designs and maneuvers Reverse engineering allows us to see what already exists. This includes any parts, structures, or processes that could benefit communities in other ways. Examining current products leads to innovation and discovery, all thanks to reverse engineering. 2. Reconstructing a product that is outdated A key part of redesigning an existing product is understanding the product itself. Reverse engineering provides the visual to work out outdated kinks in an older system. Quality is the most important aspect of this process. 3. Discovering any product vulnerabilities Similar to the previous step, reverse engineering supports finding faults in the product. This is to ensure the safety and well-being of the product's users. It is best for an issue to arise during the research phase rather than the distribution phase. 4. Bringing less expensive & more efficient products to the market Reverse engineering's main goal is to lead engineers on a path towards innovation and success. Succeeding includes lowering manufacturing costs and raising product effectiveness as much as possible. 5. Creating a reliable CAD model for future reference Most reverse engineering processes include a full-working CAD file for future references. A CAD file is created so the part can be examined digitally if future issues arise. This form of technology has enhanced engineering productivity and product expression.	
8	Explain bio mimicry in design with an example	(3)
Ans	Biomimicry is the science of applying nature-inspired designs in human engineering and invention to solve human problems. It was used to create the first flying machine, inspired by eagles and owls—this paved the way for technologies like jets and planes. It was also used in the invention of Velcro, which works in much the same way as the hooks on burrs when they stick to animal fur. By designing and producing materials, architecture, and systems that are based on biological materials and processes, we work to strike a balance with nature—to live in harmony with Mother Earth and not to continue producing global problems. Biomimicry is propelling us toward a new way of living—to sustainable assets, methods, and policies.	

	Example The head of Japan's Sanyo Shinkansen bullet train from Osaka to Hataka is modeled on the beak of the kingfisher bird.	
9	Describe ethics to be followed in engineering design	(3)
Ans	• The professional societies also undertook other kinds of activities, including promulgating design standards, and providing forums for reporting research and innovations in practice. • The professional engineering societies continue to play a leading role in setting ethical standards for designers and engineers. • These ethical standards clearly speak to the various and often conflicting obligations that an engineer must meet. • The societies also provide mechanisms for helping engineers resolve conflicting obligations, and, when asked, they provide the means for investigating and evaluating ethical behavior.	
10	Explain the significance of sustainability in engineering design	(3)
Ans	• The American Society of Mechanical Engineers (ASME) stipulates that "Engineers shall consider environmental impact in the performance of their duties." • Environmental Issues and Design • We can often characterize the environmental implications of a design in terms of the effects on air quality, water quality, energy consumption, and waste generation. • Air quality almost immediately springs to mind when we list environmental concerns related to design. • Environmentally conscious engineers should also concern themselves with issues of water quality and water consumption	

PART B

Answer any one full question from each module. Each question carries 14 marks

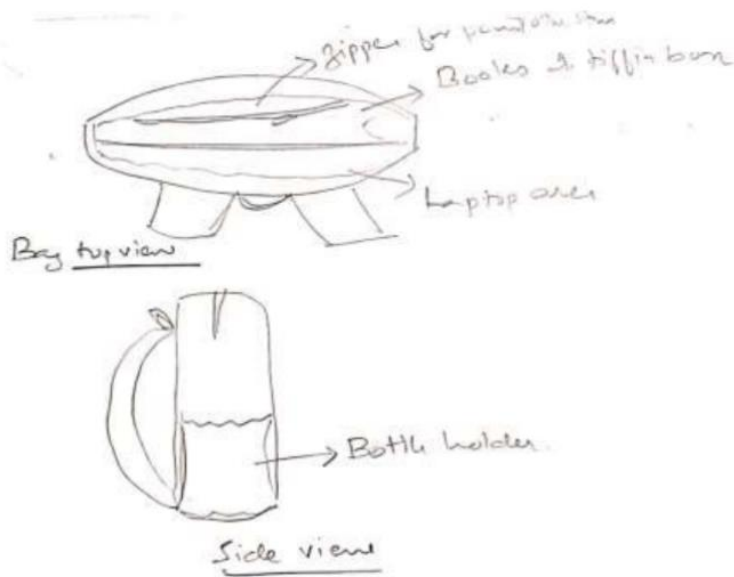
Module 1

11	Explain the design process through designing a handbag for women of age group of 15 to 25 years. Use hand sketches to support your idea	(14)
Ans	Step 1: Empathize: Research your user's need In this stage we are trying gain an empathetic understanding of the problem you're trying to solve, typically through user research. • Bag should carry books • Bag should carry tiffin box • Bag should carry water bottle • Bag should carry pen, pencil and other accessories • Zipper breaks most of the time • Shoulder got pain when carries for more time • Laptop must carry section must be there and there must be protected	

- Items get wet on a rainy day
- Bag won't last long

Step 2: Define: user's need and problems

- P1) sections must be there for books, laptop, tiffin box, pen, water bottle
 P2) Zipper and bag breaks most of the time
 P3) shoulder and back pain if carry bag for more time
 P4) Extra safety for laptop
 P5) Items get wet on a rainy day

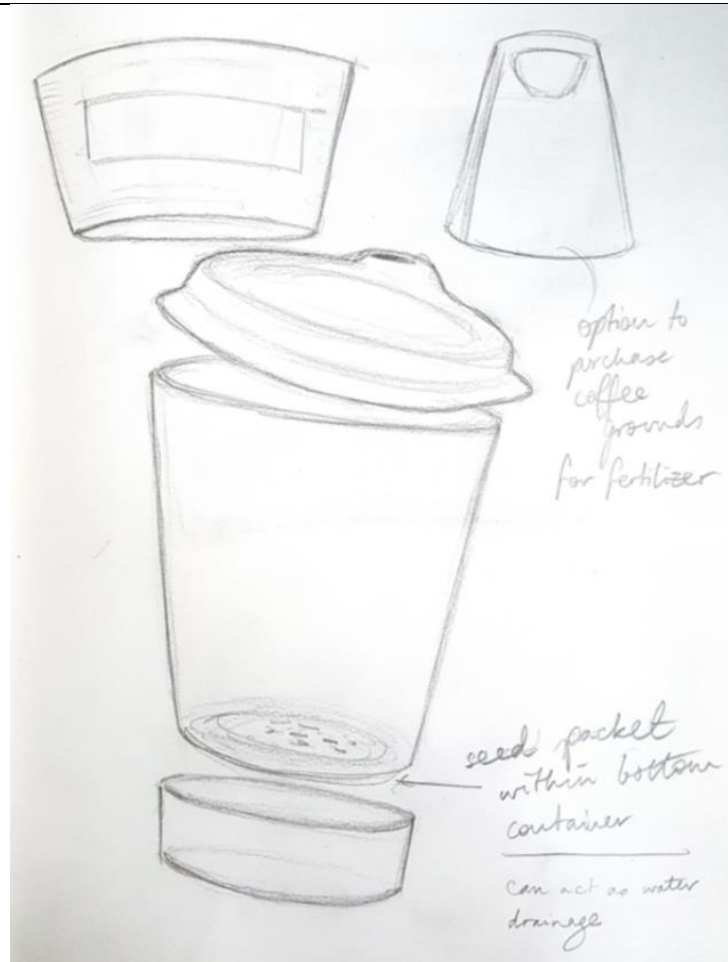
Step 3: Ideate: create idea for the problem

Better zipper material must be used like steel for long lasting

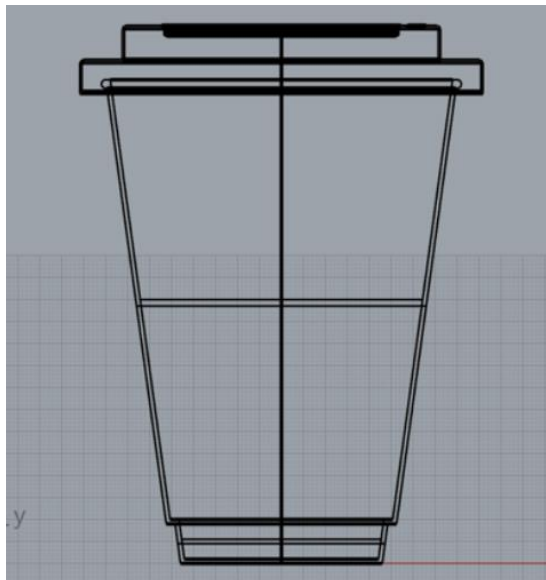
- Good and long-lasting bag material can be used
- Better stitching material can also be used
- Adding good cushions and sponge material at the shoulder strap for reduce shoulder pain
- Can reduce the back pain by adding cushion material on the back side

Step 5 : Generate Prototype

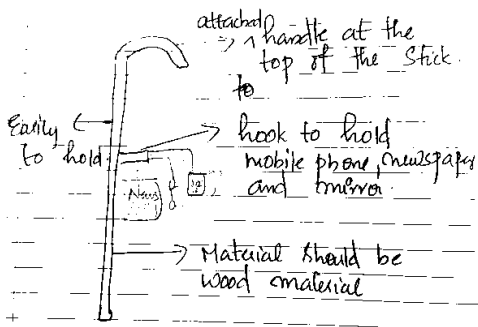
	 Step 5: Test: Try the solution In this step we can give the prototype that we made from our design to some college students and tell them to use that for some time and collect the feedback from them and from the feedback we can improve our design if we want	
12	Describe the concept of generating design alternatives and choosing a design through designing a coffee mug with the help of sketches	(14)
Ans	Explore different shapes and forms for the cup through quick sketches and spent time looking at existing cup forms and the minor variations between the different products in the market. 	



Use CAD Sketches



Module 2

13	Illustrate the design thinking process through designing a walking stick for elderly people.	(14)
Ans	To design a walking stick for elderly people. <u>Empathise</u> ⇒ Design a walking stick for elderly people. ⇒ Easy to hold. ⇒ Have maximum long-lasting quality. <u>Define</u> ⇒ Easy to hold. ⇒ Maximum long-lasting quality. ⇒ While holding the walking stick, people should feel tension free. <u>Ideate</u> ⇒ Easy to hold. ⇒ Maximum long-lasting quality. ⇒ Have an attached handle in the top of the stick to walk in a comfortable manner. <u>Prototype</u> ⇒ Elderly people should feel tension-free while holding the stick. ⇒ Have an attached handle at the top of the stick to walk in a comfortable manner. ⇒ Material should be wood material so, it will not break easily. ⇒ Easy to hold. ⇒ Have a hook at the middle part of the walking stick to store newspaper and mobile phone. <u>Test</u> ⇒ Walking stick should be efficient and long-lasting. ⇒ Easy to hold. ⇒ The people should feel tension-free. 	
14	Construct a number of possible designs and then refine them to narrow down to the best design for a drug trolley used in hospitals. Show how the divergent-convergent thinking helps in the process. Provide your rationale for each step by using hand sketches only.	(14)
	Objective: To design a drug trolley Scope/domain: Hospitals and health care centres Users: Hospital staffs	

Model 1:

Pros:

- Simple and Light weight
- Easier to keep medicines and boundaries are provided in all 3 sides which prevents from medicines falling down while moving.

Cons:

- Difficult to sort out medicines for each room.
- No doors for racks and no open table.
- Wheels have no locks



Model 2:

Pros:

- Simple and Light weight
- Easier to keep medicines

Cons:

- Difficult to sort out medicines for each room.
- No doors for racks and no open table.
- Wheels have no locks
- As there are no boundaries, there is a high chance of falling down the medicines while moving.



Model 3:

Pros:

- Simple and Light weight
- Easier to keep medicines as boundaries and lids are there for each rack.

Cons:

- Difficult to sort out medicines for each room.
- Wheels have no locks



Model 4:

Pros:

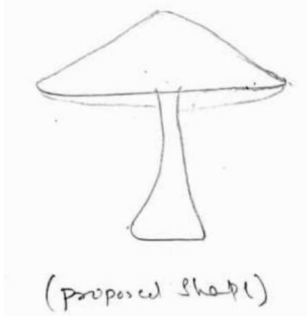
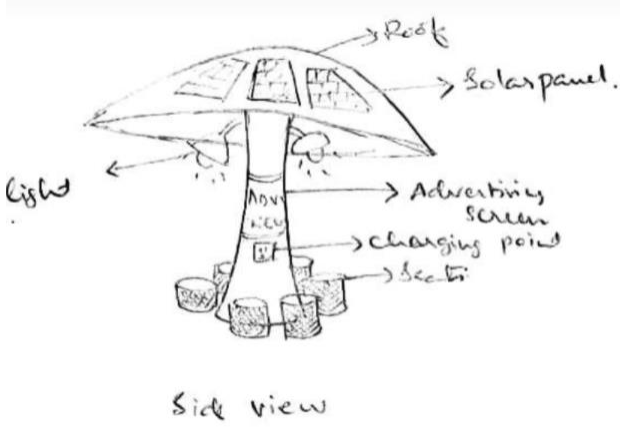
- Simple and Light weight
- Easier to keep medicines as drawers are used.
- An open table is there on the top to keep case diary/charts of patients.
- Wheels have locks

Cons:

- Difficult to sort out medicines for each room.
- No option for disposing clinical wastes.



Module 3

15	Show the development of a nature inspired design for a solar powered bus waiting shed beside a highway. Relate between natural and man-made designs. Use hand sketches to support your arguments. Solution	(14)
Ans	- Nature inspired bus waiting shed can be of many options. Some of the options are a banana leaf, or any large leaf, a big mushroom - Let us discuss the design of the bus station from one of the item, mushroom  (proposed shape) - We are using the above shape because it covers a large area and there is space underneath without affecting the rain - It has a large surface area for placing the solar panel when has sun exposure - Also its body or middle section can hold the circuit and the battery for the solar panel  Side view	
16	Describe the role of mathematical modelling in design engineering citing an example	(14)

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	• MATHEMATICAL MODELS are central to design because we have to be able to predict the behavior of the devices or systems that we are designing. • It is important for us to ask: How do we create mathematical models? How do we validate such models? How do we use them? And, are there any limits on their use? • Basic Principles of Mathematical Modeling • Why do we need a model? • For what will we use the model? • What do we want to find with this model? • What data are we given? • What can we assume? • How should we develop this model, that is, what are the appropriate physical principles we need to apply? • What will our model predict? • Can we verify the model's predictions (i.e., are our calculations correct?) • Are the predictions valid (i.e., do our predictions conform to what we observe?) • Can we improve the model?	
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Module 4

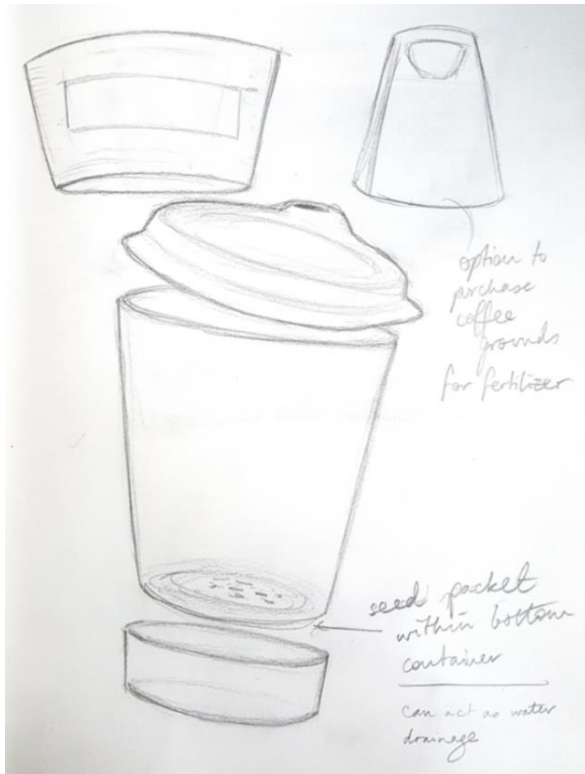
17	Illustrate modular design approach for designing of desktop computers Modular design, or modularity in design, is a design principle that subdivides a system into smaller parts called <i>modules</i> (such as modular process skids), which can be independently created, modified, replaced, or exchanged with other modules or between different systems. A modular design can be characterized by functional partitioning into discrete scalable and reusable modules, rigorous use of well-defined modular interfaces, and making use of industry standards for interfaces. In this context modularity is at the component level, and has a single dimension, component slottability. A modular system with this limited modularity is generally known as a platform system that uses modular components. Examples are car platforms or the USB port in computer engineering platforms. In design theory this is distinct from a modular system which has higher dimensional modularity and degrees of freedom. A modular system design has no distinct lifetime and exhibits flexibility in at least three	(14)
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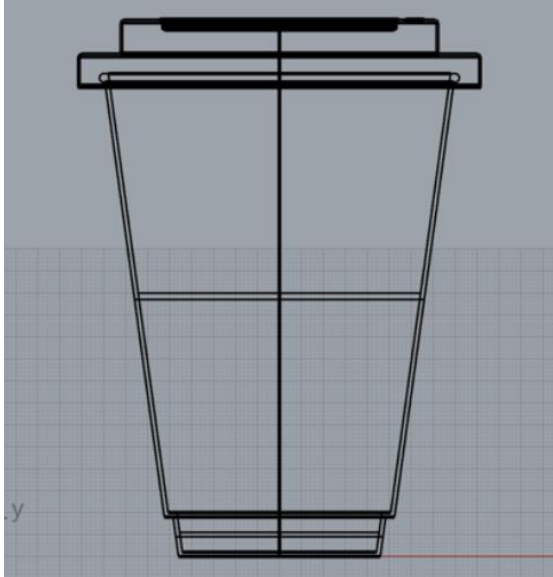
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1	dimensions. In this respect modular systems are very rare in markets. Mero architectural systems are the closest example to a modular system in terms of hard products in markets. Weapons platforms, especially in aerospace, tend to be modular systems, wherein the airframe is designed to be upgraded multiple times during its lifetime, without the purchase of a completely new system. Modularity is best defined by the dimensions effected or the degrees of freedom in form, cost, or operation. Modularity offers benefits such as reduction in cost (customization can be limited to a portion of the system, rather than needing an overhaul of the entire system), interoperability, shorter learning time, flexibility in design, non-generationally constrained augmentation or updating (adding new solution by merely plugging in a new module), and exclusion. Modularity in platform systems, offer benefits in returning margins to scale, reduced product development cost, reduced O&M costs, and time to market. Platform systems have enabled the wide use of system design in markets and the ability for product companies to separate the rate of the product cycle from the R&D paths. The biggest drawback with modular systems is the designer or engineer. Most designers are poorly trained in systems analysis and most engineers are poorly trained in design.	
18	Demonstrate the concept of ergonomics through design of a table lamp	(14)
	So for this design I am actually going to focus on aesthetics. I said I would do this on the coffee table but I soon realized that it would make more sense for the coffee table, reclining chair, and couch to all match. I want this lamp to match as well but I want to make it stand out. The whole purpose of this project was to create an ergonomic furniture set. Not all of the furniture pieces are completely ergonomic when they are by themselves but when they are put together they make an ergonomic living room that utilizes the function of the living room as well as the space.	

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Module 5

17	Illustrate the changes in design of disposable tea cup in terms of production, use and sustainability	(14)
	Explore different shapes and forms for the cup through quick sketches and spent time looking at existing cup forms and the minor variations between the different products in the market.   Use CAD Sketches	

		
20	Describe the how to estimate the cost of a website design stage? Show how economics will influence the engineering designs.	(14)
	Mainly for a product design there are three costs 1) labour cost , 2) material cost and 3) overhead cost. Let us discuss each 1) Labour cost • The labour cost of the website creation is mainly for the website designers and programmers. For the creation of the established shopping website fully we need minimum 10 to 15 designers, programmers and supporting engineers • The estimated time for the full functioning of the website may take around 5- 8 months • We need to pay the salary for the workers at that time period. Let us take nearly 15000 per month one person. Then the total salary for a month will be $15 \times 15000 = 225000$. • That amount should provide for the estimated time period of 8 months. Which will cost $8 \times 225000 = 18,00,000$ rs • After the website creation we can provide support and annual maintenance of the website afterwards. This may cost 3-5 worker's salary and other benefits. The salary may cost $5 \times 15000 = 75000$ rs. This cost will continue as long as the website is functioning. • The other works like marketing and social media marketing can be done by 2 workers. The salary for them may cost $2 \times 10000 = 200000$ rs 2) material cost • We can also reduce the cost by hiring a website creation outsourcing company for to do the jobs, which will reduce the employee salary and other expences Material cost • The main material cost for the website creation is computer	

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	expense. At the time of the website creation above mentioned workers need 10 – 15 computer sets for the website design and developing. So for a single computer set will cost nearly 30000 per system. So the total cost will be $15 \times 30000 = 450000$. • Also, must have servers/ cloud storage for the smooth functioning and controlling of the website which will cost 100000 per server • If we chose the cloud storage then we must pay yearly rent for the cloud servers • There must be an office for the functioning of the website we can have own building or rented building. The cost will be depending on the option • The initial cost of this will be high for a starting company. We can also reduce this cost by outsourcing the work with another firm • The other cost will be website domain cost which will be nearly 3000/year which is a mandatory cost in each year • Another mandatory cost is website hosting which nearly cost 2000/year • Some of the costs can be reduced by outsourcing the works to another firms. 3) Overhead costs • Over head cost are the cost happening after the cost estimation. There will be always overflow of the cost estimation after the establishment and while the functioning of the website. We must consider those options also. Some of the overhead cost is mentioned below - o Some cases we might need to develop an app for the website to use it in mobile phones. This may cost additional - o Some cases to improve the traffic and working of the website we might need additional certificate for website like ssl certificate. - o Additional publicity cost will be there for reaching the website to more people. This can be done by social media advertising, additional discounts for the products. These steps can cost extra	
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